

ESAT CAMP MOCK B

Full sitting

EC-FULL-B

MATHEMATICS 1

40 MINUTES

MATHEMATICS 2

40 MINUTES

PHYSICS

40 MINUTES

CHEMISTRY

40 MINUTES

BIOLOGY

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin**.

This paper contains **5 modules** (Mathematics 1, Mathematics 2, Physics, Chemistry, Biology). Each module has **27 multiple-choice questions** and is designed for **40 minutes** (about **200 minutes** in total if you sit every module).

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.

ESAT CAMP · Independent preparation resource

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Mathematics 1

- 1 A shop sells two items, item P and item Q.

In January, the price of item P was 25% higher than the price of item Q.

In February, the price of item P decreased by 20%, while the price of item Q increased by 10%.

In March, the price of item P increased by 15%, while the price of item Q decreased by 5%.

In March, the price of item P was £6.30 more than the price of item Q.

What was the price of item Q in January?

- A £42
- B £50
- C £60
- D £66
- E £70
- F £75
- G £600

- 2 The expression $\frac{3\sqrt{3} + 3}{\sqrt{3} - 1}$ can be written in the form $a + b\sqrt{3}$, where a and b are integers.

What is the value of $a + b$?

- A 0
- B 3
- C 6
- D 9
- E 18

- 3 A fair four-sided die has faces labelled 1, 2, 3, 4.
A fair six-sided die has faces labelled 1, 1, 2, 3, 4, 6.

Both dice are rolled simultaneously. Let a be the score on the four-sided die and b be the score on the six-sided die.

What is the probability that the quadratic equation

$$x^2 + ax + b = 0$$

has real roots?

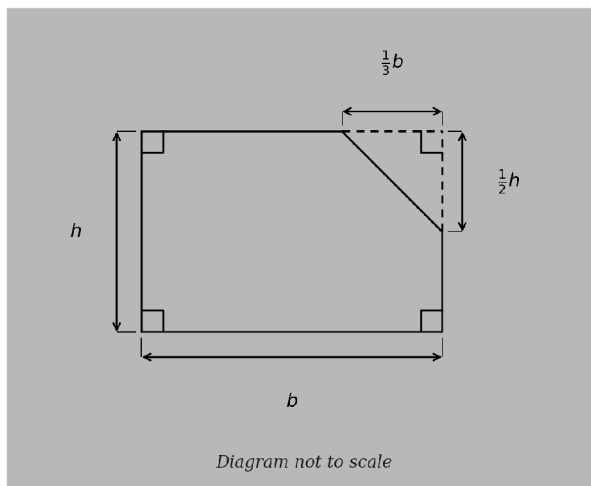
- A $\frac{7}{24}$
- B $\frac{3}{8}$
- C $\frac{5}{12}$
- D $\frac{11}{24}$
- E $\frac{7}{12}$

- 4 A square of side length x has area A .

The square is modified to form the pentagon shown in the diagram below:

- the base length b is 50% greater than x ,
- the height h is 20% less than x ,
- a right-angled triangular corner with base $\frac{1}{3}b$ and height $\frac{1}{2}h$ is removed from the top-right corner.

What is the area of the resulting pentagon in terms of A ?

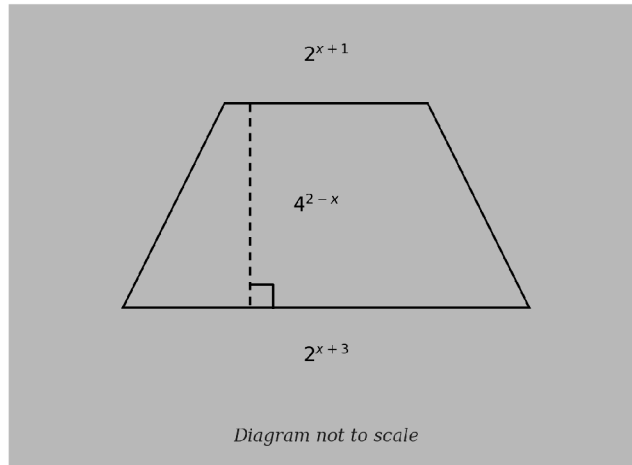


- A $\frac{9}{10}A$
- B A
- C $\frac{11}{10}A$
- D $\frac{67}{60}A$
- E $\frac{6}{5}A$
- F $\frac{13}{10}A$
- 5 A component has a volume of V cubic inches. One inch is equivalent to L centimetres. The component is made from a material with a density of D g/cm³. The material is supplied in blocks, each with a mass of M kg. Each block costs C pounds.
- Which of the following expressions gives the cost of the material for the component, in pence?
(There are 100 pence in one pound and 1000 grams in one kilogram.)

- A $\frac{VCDL^3}{10M}$
- B $\frac{VCDL}{10M}$
- C $\frac{VCD}{10ML^3}$
- D $\frac{100VCDL^3}{M}$
- E $\frac{VCDL^3}{1000M}$
- F $\frac{VCL^3}{10MD}$

- 6 The diagram shows a trapezium with parallel horizontal sides of length 2^{x+1} and 2^{x+3} , and perpendicular height 4^{2-x} .

Given that the area of the trapezium is $40\sqrt{2}$, what is the value of x ?



- A -1.5
 B -0.5
 C 0.5
 D 1.5
 E 2.5
- 7 Let x and y be positive real numbers such that:

$$u = \log_2 \left(\frac{x^3 y^2}{32} \right)$$

and

$$v = \log_2 \left(\frac{xy^2}{8} \right)$$

Given that $u - v = 4$, what is the value of x ?

- A $\frac{1}{2}$
 B 2
 C 3
 D 4
 E 8

- 8 Two solid objects, X and Y , are mathematically similar. The volume of X is 24 and the volume of Y is 81. The total surface area of X is S .

Which expression gives the total surface area of Y ?

- A $\frac{8}{27}S$
- B $\frac{4}{9}S$
- C $\frac{2}{3}S$
- D $\frac{3}{2}S$
- E $\frac{9}{4}S$
- F $\frac{27}{8}S$

- 9 A box contains only black discs and white discs. The total number of discs in the box is 15.

There are more black discs than white discs in the box.

Two discs are drawn at random from the box without replacement.

The probability that both discs drawn are of the same colour is $\frac{11}{21}$.

What is the probability that both discs drawn are black?

- A $\frac{2}{21}$
- B $\frac{1}{7}$
- C $\frac{5}{21}$
- D $\frac{2}{7}$
- E $\frac{1}{3}$
- F $\frac{3}{7}$
- G $\frac{4}{9}$
- H $\frac{2}{3}$

- 10 A set of 5 distinct integers has a mean of 10, a median of 5 and a range of 20.

The integer 8 is added to the set.

Which of the following correctly describes the changes to the mean, the median and the range of the set?

- A Mean decreases, Median unchanged, Range increases
- B Mean increases, Median increases, Range unchanged
- C Mean decreases, Median increases, Range unchanged
- D Mean increases, Median unchanged, Range increases
- E Mean decreases, Median increases, Range increases

- 11 An unlimited supply of gold and silver coins is available. The probability of selecting a gold coin is $\frac{1}{3}$.

Two boxes, Box 1 and Box 2, are prepared. To prepare each box, two coins are selected from the supply and placed inside.

It is known that each box contains at least one gold coin.

One coin is then drawn from Box 1 and one coin is drawn from Box 2.

What is the probability that both coins drawn are gold?

- A $\frac{1}{25}$
- B $\frac{1}{9}$
- C $\frac{9}{25}$
- D $\frac{3}{5}$
- E $\frac{9}{16}$
- F $\frac{4}{5}$

- 12 A factory processes 1200 kg of raw material in batches of 20 kg.

In Stage 1, each batch is converted into 10 units of substance P and 5 units of substance G.

In Stage 2, all of substance P is processed, with each unit of P producing 0.5 units of substance G.

All the substance G from both stages is used to create a final product, F. 3 units of G are required to produce 2 units of F. The product F is produced at a constant rate of 5 units per minute.

What is the total time, in minutes, taken to produce all of F?

- A 40
- B 80
- C 90
- D 120
- E 180
- F 240

- 13** A composite cylindrical rod of length L consists of a solid inner core of radius R and a surrounding outer shell of inner radius R and outer radius $2R$.

The core is made of a material with density ρ and has mass M .

The shell is made of a material with density 2ρ .

What is the total mass of the rod?

- A** $3M$
- B** $4M$
- C** $6M$
- D** $7M$
- E** $8M$
- F** $9M$

- 14** A set of n distinct items ($n \geq 3$) is arranged in a random linear order. Two specific items, X and Y , are chosen from the set.

Which expression gives the probability that item X appears earlier in the arrangement than item Y ?

- A** $\frac{1}{n(n-1)}$
- B** $\frac{1}{n}$
- C** $\frac{2}{n}$
- D** $\frac{1}{2}$
- E** $\frac{n-1}{n}$

- 15** The quadratic equation in x

$$x^2 + (4^{2-2k} \times 8^k)x + 16^{2-k} = 0$$

has exactly one real root.

What is the value of k ?

- A** -3
- B** -2
- C** 0
- D** 1
- E** 2
- F** 3

- 16** Five years ago, the ratio of the age of P to the age of Q was 5 : 2. In 12 years, P will be 21 years older than Q.
What is the current age of P?
- A** 19
B 23
C 35
D 40
E 47
- 17** A 3D printer extrudes a cylindrical filament of plastic with a diameter of 4 mm. The filament is fed at a constant rate of 5 mm per second for a total of 10 minutes.
The density of the plastic is $\frac{2.5}{\pi}$ g/cm³.
What is the total mass of plastic used, in grams?
- A** 0.5
B 2
C 30
D 120
E 300
F 3000
- 18** An exam paper is formatted in sections. For any integer $k \geq 1$, Section k contains exactly k pages.
The paper is printed such that every section must begin on an odd-numbered page. To achieve this, if a section ends on an odd-numbered page, a single blank page is inserted immediately after it. Section 1 begins on page 1.
What is the page number on which Section 51 begins?
- A** 1275
B 1276
C 1300
D 1301
E 1302
F 1326

- 19 A region in the xy -plane is defined by the following inequalities:

$$0 \leq x \leq a$$

$$0 \leq y \leq a$$

$$x^2 + y^2 \geq a^2$$

where a is a positive constant.

What is the area of this region?

A $a^2(1 - \frac{\pi}{4})$

B $\frac{\pi a^2}{4}$

C a^2

D $a^2(1 - \frac{\pi}{2})$

E $a^2(1 - \pi)$

F $\frac{\pi a^2}{2}$

- 20 An equilateral triangle and a regular 12-sided polygon are inscribed in the same circle. The area of the equilateral triangle is $27\sqrt{3}$.

What is the area of the regular 12-sided polygon?

A 27

B 54

C 108

D $108\sqrt{3}$

E 216

F 324

- 21 A cubic polynomial $P(x)$ with leading coefficient 1 can be factorised as $P(x) = (x - r_1)(x - r_2)(x - r_3)$, where r_1, r_2 and r_3 are positive integers.

The curve $y = P(x)$ passes through the point $(3, -4)$.

How many distinct polynomials $P(x)$ satisfy these conditions?

A 1

B 3

C 4

D 5

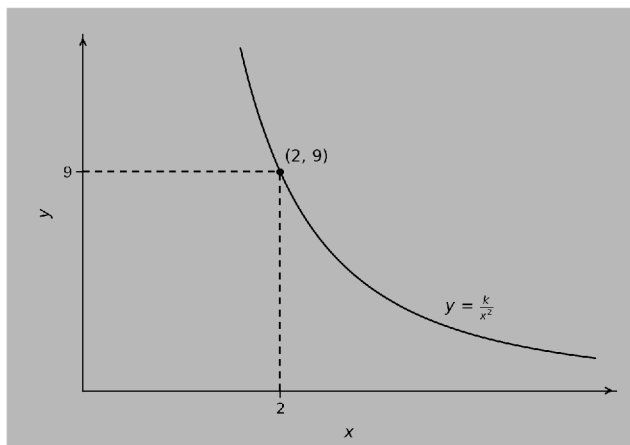
E 6

F 18

- 22** The diagram shows part of the curve $y = \frac{k}{x^2}$, where k is a positive constant and $x > 0$. The curve passes through the point $(2, 9)$.

Also, x is directly proportional to z^3 . When $z = 2$, $x = 24$.

Which of the following is an expression for y in terms of z ?



- A** $y = \frac{1}{4z^6}$
- B** $y = \frac{4}{z^6}$
- C** $y = \frac{4}{z^5}$
- D** $y = \frac{6}{z^3}$
- E** $y = \frac{12}{z^6}$
- F** $y = \frac{36}{z^6}$
- G** $y = \frac{81z^6}{4}$
- H** $y = 4z^6$
- 23** A system's success depends on two independent paths, A and B. The system succeeds if at least one path succeeds. Each path consists of 3 relays in series. A path succeeds only if all 3 of its relays succeed. All relays are identical and independent, each with a probability p of success. In 128 independent trials, the expected number of system successes is 30.

What is the value of p ?

- A** $\frac{1}{24}$
- B** $\frac{1}{8}$
- C** $\frac{3}{8}$
- D** $\frac{1}{2}$
- E** $\frac{5}{8}$
- F** $\frac{7}{8}$

- 24** Three regular polygons meet at a single vertex, completely covering the angle around it. One polygon is a square. The other two have n sides and m sides respectively, where $n < m$. What is the maximum possible value of m ?
- A** 8
 - B** 12
 - C** 16
 - D** 20
 - E** 24
 - F** 42
- 25** Three quantities x , y and z are related by the ratios $x : y = 3 : 2$ and $z : y = 5 : 4$. What is the ratio $x : z$?
- A** 3 : 5
 - B** 5 : 6
 - C** 6 : 5
 - D** 8 : 15
 - E** 15 : 8
- 26** The variables x and y are positive.
- For which of the following relationships is y inversely proportional to x ?
- A** The sum $x + y$ is a constant.
 - B** $y = \frac{k}{x^2}$ for some non-zero constant k .
 - C** $y = \frac{k}{x + c}$ for some non-zero constants k and c .
 - D** The product xy is a non-zero constant.
 - E** The ratio $\frac{y}{x}$ is a non-zero constant.
 - F** $y = \frac{k}{x} + c$ for some non-zero constants k and c .

27 The following four lengths are denoted by P, Q, R, and S.

$$P = 5\mu\text{m}$$

$$Q = 0.02\text{mm}$$

$$R = 200\text{nm}$$

$$S = 5 \times 10^{-5}\text{m}$$

Arrange these lengths in order, from smallest to largest.

A R, P, Q, S

B S, Q, P, R

C P, Q, S, R

D Q, S, R, P

E R, P, S, Q

F R, Q, S, P

Mathematics 2

- 1 What is the value of the definite integral

$$\int_{-2}^2 (3x^2 + 5 + x^3 \cos x) dx$$

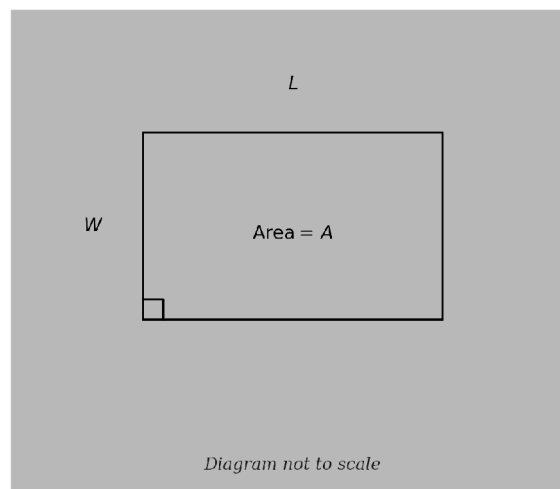
- A 0
- B 16
- C 18
- D 36
- E 72

- 2 The diagram shows a rectangle of length L , width W , and area A .

Given that

$$A = \frac{3x^2 - 11x - 4}{9x^2 - 1} \quad \text{and} \quad W = \frac{x^2 - 2x - 8}{3x^2 + 14x - 5}$$

which one of the following is an expression for L in its simplest form?



- A $\frac{x-5}{x+2}$
- B $\frac{x-5}{x-2}$
- C $\frac{x+2}{x+5}$
- D $\frac{x+5}{x-2}$
- E $\frac{x+5}{x+2}$

- 3 The function $f(x)$ is continuous on the interval $0 \leq x \leq 8$. The curve $y = f(x)$ crosses the x -axis exactly once in this interval.

Given that:

$$\int_0^8 f(x) dx = 2$$

and the total area of the region bounded by the curve $y = f(x)$, the x -axis, and the lines $x = 0$ and $x = 8$ is 12.

What is the area of the region lying below the x -axis?

- A 2
B 5
C 6
D 7
E 10
- 4 The table shows the number of items and the mean score for three groups of data, A, B, and C.

Group	Number of items	Mean score
A	n	12
B	$2n$	18
C	k	30

When all three groups are combined, the overall mean score of all the items is 20.

What is the ratio $k : n$?

- 5 A curve has equation $y = f(x)$. The derivative of the function is given by $f'(x) = 3x^2 - 12$. The distance between the two stationary points of the curve is $4\sqrt{65}$.

What is the y -intercept of the curve?

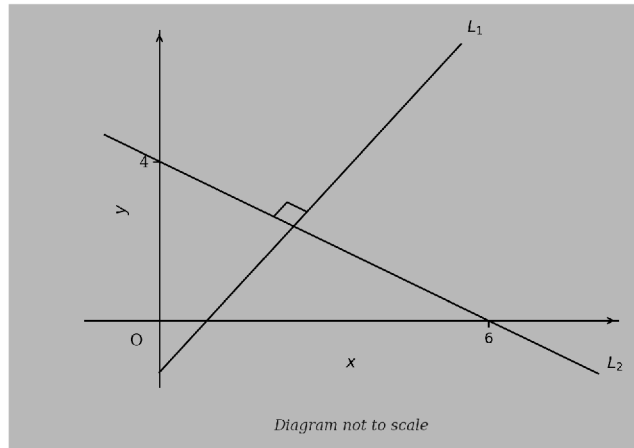
- A -16
B 0
C 16
D 32
E Cannot be determined

- 6 The diagram shows two perpendicular straight lines, L_1 and L_2 , in the Cartesian plane.

Line L_1 passes through the points with coordinates $(2 - k, 3k - 1)$ and $(k + 4, k + 5)$, where k is a constant.

Line L_2 intersects the coordinate axes at $(0, 4)$ and $(6, 0)$.

What is the value of k ?



- A -9
- B $-\frac{3}{5}$
- C $\frac{3}{5}$
- D $\frac{7}{5}$
- E $\frac{5}{3}$
- F 11
- 7 What is the sum of all real values of x that satisfy the equation $\sin(2x) = \cos(x)$ in the interval $0 \leq x \leq \pi$?
- A $\frac{\pi}{2}$
- B $\frac{2\pi}{3}$
- C π
- D $\frac{4\pi}{3}$
- E $\frac{3\pi}{2}$

- 8 A rhombus has diagonals of length $2\sqrt{5}$ cm and 4 cm. An enlargement of the rhombus has an area of $80\sqrt{5}$ cm².

What is the perimeter of the enlarged rhombus?

- A $6\sqrt{5}$ cm
- B $12\sqrt{5}$ cm
- C $24\sqrt{5}$ cm
- D $48\sqrt{5}$ cm
- E 240 cm
- F 480 cm

- 9 Given that $\sin(\theta) = \frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\cot(2\theta)$.

- A $\frac{24}{7}$
- B $\frac{7}{24}$
- C $-\frac{7}{24}$
- D $-\frac{24}{7}$
- E $-\frac{7}{12}$
- F $-\frac{32}{21}$

- 10 The positive variables u , v , w , and x satisfy the following relationships:

- u is inversely proportional to the square root of v
- v is directly proportional to the cube of w
- w is inversely proportional to the square of x

Which of the following correctly describes the relationship between u and x ?

- A u is directly proportional to x^3
- B u is inversely proportional to x^3
- C u is directly proportional to x^6
- D u is inversely proportional to x^6
- E u is directly proportional to $x^{3/2}$
- F u is inversely proportional to $x^{3/2}$

- 11** A geometric series has a common ratio r . The value of r is the result of increasing the number 1 by $p\%$, and then decreasing the new value by $p\%$, where $p > 0$.
- The sum to infinity of this series exists.
- Which of the following gives the complete set of possible values for p ?
- A** $0 < p < 100$
 - B** $0 < p < 100\sqrt{2}$
 - C** $p > 0$
 - D** $p > 100\sqrt{2}$
 - E** $0 < p < 200$
 - F** No values of p .
- 12** The expression $(3x + 2)^6$ is expanded in ascending powers of x .
- How many of the coefficients in this expansion are divisible by 6?
- A** 4
 - B** 5
 - C** 6
 - D** 7
 - E** 2
- 13** The function $f(x) = ax^2 + bx + c$ has a stationary point at $x = 2$. The gradient of the curve $y = f(x)$ at $x = 0$ is 12. Given that $f(1) = 15$, what is the value of c ?
- A** 0
 - B** 6
 - C** 9
 - D** 12
 - E** 24
- 14** A circle has the equation
- $$x^2 + y^2 - 2x + 2y - 23 = 0$$
- Find the area of the triangle bounded by the coordinate axes and the tangent to the circle at the point $(4, 3)$.
- A** 6
 - B** 24
 - C** 27
 - D** 48
 - E** $\frac{49}{24}$

- 15 The function $f(x)$ is defined for $0 \leq x \leq 6$ by:

$$f(x) = \begin{cases} 11 + 2x - x^2 & \text{for } 0 \leq x \leq 3 \\ 8x - x^2 - 6 & \text{for } 3 < x \leq 6 \end{cases}$$

What is the maximum value of $f(x)$?

- A 9
 - B 10
 - C 11
 - D 12
- 16 A function f is defined by

$$f(x) = \log_3 \left(\frac{ax + b}{x} \right)$$

where a and b are constants.

Given that $f(1) = 2$ and $f(4) = 1$, find the value of x such that $f(x) = -1$.

- A -12
 - B -6
 - C -4
 - D -2
 - E 4
- 17 How many solutions does the equation $2 \cos \left(2x + \frac{\pi}{3} \right) = 1$ have in the interval $0 \leq x \leq 2\pi$?
- A 2
 - B 3
 - C 4
 - D 5
 - E 6

- 18 Quantities X and Y change in discrete steps $n = 0, 1, 2, \dots$. Initially, $X_0 = 20Y_0$, where $Y_0 > 0$. In each step, X decreases by 60% of its value at the start of the step, and Y decreases by 20% of its value at the start of the step. What is the minimum value of n such that $X_n < Y_n$?

- A 2
- B 4
- C 5
- D 6

- 19** The curve with equation $y = 4^x - 10 \cdot 2^x$ intersects the horizontal line $y = c$ at two distinct points. The x -coordinates of these points are p and q , where $q > p$. Given that $q - p = 2$, what is the value of the constant c ?
- A** -25
B -24
C -16
D -4
E 16
- 20** A circle passes through the points $(2, 1)$ and $(4, 5)$ and is tangent to the x -axis. What is the largest possible radius of the circle?
- A** 1
B 2.5
C 3
D 5
E $5\sqrt{2}$
- 21** Let n be a positive odd integer. Two distinct coefficients in the expansion of $(1 + x)^n$ are chosen at random. The probability that these two coefficients are equal is $\frac{1}{23}$. What is the value of n ?
- A** 21
B 22
C 23
D 24
E 25
F 45
- 22** A sequence of three positive integers has a sum of 6 and a sum of the squares of its terms of 18. How many such distinct sequences are there?
- A** 0
B 1
C 2
D 3
E 6
F 9

- 23** A convergent geometric series has first term $\tan \theta$ and third term $\cot \theta$. The second term of the series is negative, and the sum to infinity is 1.

What is the value of the first term?

- A** $\frac{1}{2}$
- B** 1
- C** $\frac{1+\sqrt{5}}{2}$
- D** $\frac{1-\sqrt{5}}{2}$
- E** $\frac{-1+\sqrt{5}}{2}$

- 24** A particle travels at a constant speed v from the point $(-2R, 0)$ to the point $(2R, 0)$, where $R > 0$.

The particle's path is chosen to be the shortest possible route that does not enter the region $x^2 + y^2 < R^2$.

What is the minimum time for this journey?

- A** $\frac{R}{v}(2\sqrt{3} + \frac{\pi}{3})$
- B** $\frac{R}{v}(\pi + 2)$
- C** $\frac{R}{v}(2\sqrt{3} + \frac{2\pi}{3})$
- D** $\frac{R}{v}(2\sqrt{5})$
- E** $\frac{R}{v}(2\sqrt{3})$

- 25** A function f is defined by

$$f(x) = \frac{x^2 + 5x + 4}{x^2 + 7x + 12} \div \frac{x + 1}{3x + 9}$$

for all values of x for which the expression is defined.

What is the value of $f'(4)$?

- A** 0
- B** $\frac{1}{3}$
- C** 1
- D** 3
- E** 4
- F** -3

- 26** A geometric progression has first term a and common ratio r .
The first term is

$$a = \frac{1}{x+2}$$

and the sum to infinity is

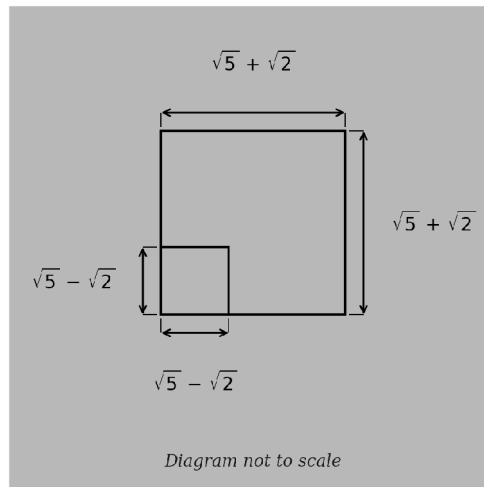
$$S_{\infty} = \frac{x-1}{x^2-4}$$

What is the common ratio r ?

- A** $\frac{1}{x-1}$
- B** $\frac{x-2}{x-1}$
- C** $\frac{1}{2-x}$
- D** $\frac{2x-3}{x-1}$
- E** $-\frac{1}{x-1}$
- F** $\frac{x+1}{x+2}$

- 27** The diagram shows a square of side length $\sqrt{5} + \sqrt{2}$. A smaller square of side length $\sqrt{5} - \sqrt{2}$ is removed from one corner, leaving the shaded region.

What is the area of the shaded region?



- A** 4
- B** 8
- C** 14
- D** $2\sqrt{10}$
- E** $4\sqrt{10}$
- F** $7 + 2\sqrt{10}$
- G** $14 - 4\sqrt{10}$
- H** $14 + 4\sqrt{10}$

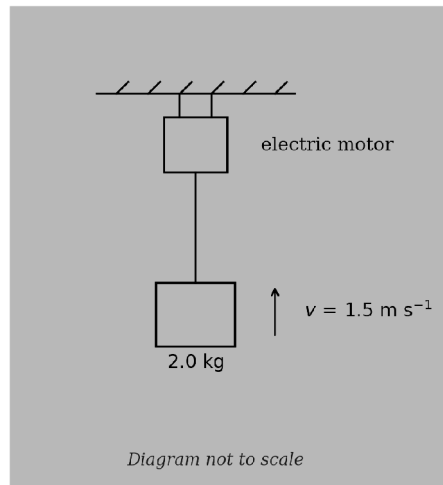
Physics

- 1 An electric motor is used to lift a block of mass 2.0 kg vertically at a constant speed of 1.5 m s^{-1} , as shown in the diagram.

The electrical power supplied to the motor is 50 W .

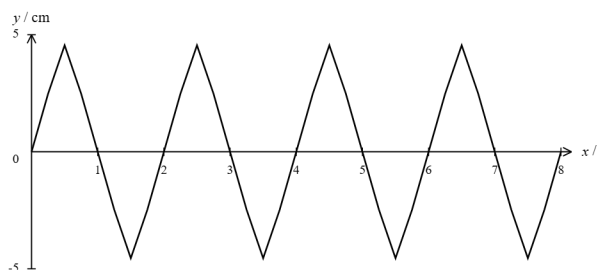
How much energy is wasted by the motor in 5.0 minutes?

[gravitational field strength = 10 N kg^{-1}]



- A 100 J
 B 150 J
 C 6000 J
 D 9000 J
 E 15 000 J
- 2 A stationary neutral atom contains a total of N constituent particles (protons, neutrons and electrons).
 The atom undergoes alpha decay, and the resulting daughter product is also a neutral atom.
 What is the total number of constituent particles in the daughter atom?
- A $N - 8$
 B $N - 6$
 C $N - 4$
 D $N - 2$
 E N
 F $N + 1$

- 3 The graph shows a snapshot of a transverse wave on a string, with displacement y in cm and position x in m. The wave travels at a speed of 10 m s^{-1} .



What is the average speed of a particle on the string during one complete oscillation?

- A 0.50 m s^{-1}
 - B 1.0 m s^{-1}
 - C 2.0 m s^{-1}
 - D 4.0 m s^{-1}
 - E 10 m s^{-1}
 - F $\pi \text{ m s}^{-1}$
- 4 A vehicle of constant mass m accelerates from rest along a horizontal straight track. The engine provides a constant mechanical power output P , and all resistive forces are negligible.
- What is happening to the magnitude of the acceleration of the vehicle as time increases?
- A It is increasing at an increasing rate.
 - B It is increasing at a constant rate.
 - C It is increasing at a decreasing rate.
 - D It is not changing.
 - E It is decreasing at an increasing rate.
 - F It is decreasing at a constant rate.
 - G It is decreasing at a decreasing rate.

- 5 In a mass spectrometer, the radius of a nucleus's circular path is proportional to its mass-to-charge ratio. An unknown nucleus is found to have a mass-to-charge ratio that is 2.5 times that of a proton.

Assuming a nucleus's mass is proportional to its mass number A and its charge is proportional to its atomic number Z , which of the following could be the unknown nucleus? The nuclei are described by (number of protons, number of neutrons).

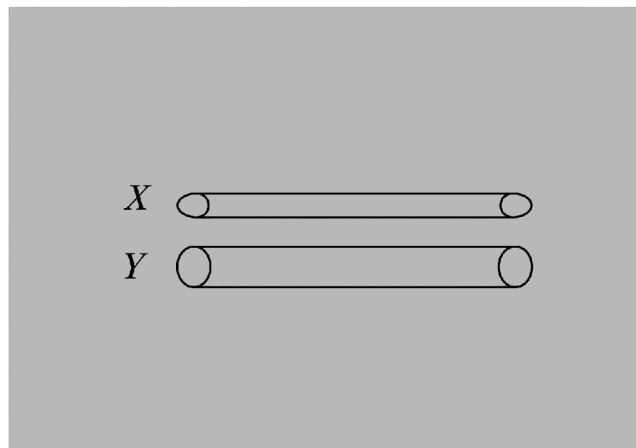
- A (2, 2)
- B (2, 3)
- C (2, 4)
- D (2, 5)
- E (3, 2)
- F (5, 2)

- 6 Two wires, X and Y, are made from the same metal.

Wire Y has three times the length and twice the diameter of wire X.

The electrical resistance of wire X is R .

What is the electrical resistance of wire Y?



concept image

- A $\frac{3}{4}R$
- B $\frac{3}{2}R$
- C $\frac{4}{3}R$
- D $\frac{9}{4}R$
- E $6R$
- F $12R$

- 7 The graph of terminal potential difference, V , against current, I , for a power source is a straight line.

The line passes through the point where $I = 1.0\text{ A}$ and $V = 10.0\text{ V}$, and the point where $I = 4.0\text{ A}$ and $V = 4.0\text{ V}$.

What is the internal resistance of the power source?

- A $0.50\ \Omega$
- B $1.0\ \Omega$
- C $2.0\ \Omega$
- D $3.0\ \Omega$
- E $6.0\ \Omega$
- F $10.0\ \Omega$

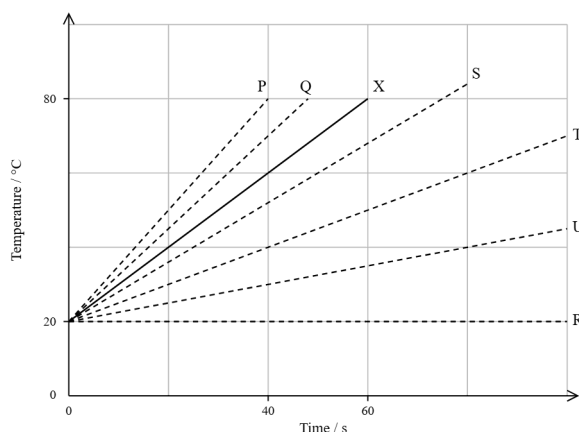
- 8 A rock sample is known to have initially consisted entirely of a radioactive parent isotope P. This isotope decays to a single stable daughter isotope D. At a time t , the ratio of the number of atoms of D to the number of atoms of P is 3 : 1.

What is the ratio of the number of atoms of D to the number of atoms of P at time $2t$?

- A 6 : 1
- B 7 : 1
- C 9 : 1
- D 15 : 1
- E 31 : 1
- F 63 : 1

- 9** An electric heating element consists of three identical resistors connected in series to a power supply of constant voltage. The element is used to heat a fixed mass of water, starting from an initial temperature of 20°C .

The solid line, Graph X, in the diagram shows the temperature of the water as a function of time. No heat is lost to the surroundings.



One of the resistors fails and becomes a short circuit. The experiment is repeated with the same mass of water from the same initial temperature.

Which of the dashed lines (P, Q, R, S, T, or U) represents the new temperature–time graph?

- A** P
B Q
C R
D S
E T
F U
- 10** Two electric heaters, with power ratings of 150 W and 100 W, are used in an experiment. Each heater is used to raise the temperature of an identical mass of a liquid by the same temperature difference.
- The 150 W heater completes this process 60 s faster than the 100 W heater.
- Assuming no energy is lost to the surroundings, what is the total thermal energy transferred to the liquid?
- A** 3.0 kJ
B 3.6 kJ
C 6.0 kJ
D 9.0 kJ
E 18 kJ

- 11 A particle moves along a straight line such that its velocity v (in m s^{-1}) at time $t \geq 0$ (in s) is given by

$$v = t^3 - 5t^2 + 8t - 4$$

What is the complete set of times t for which the particle is moving in the positive direction?

- A $t > 1$
- B $t > 2$
- C $1 < t < 2$
- D $1 < t < 2$ or $t > 2$
- E $0 \leq t < 1$ or $t > 2$
- F $t < 1$ or $t > 2$

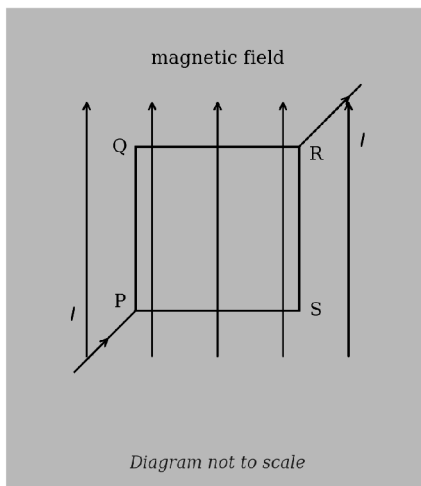
- 12** A rigid square frame $PQRS$, made from uniform conducting wire of constant cross-section and resistance per unit length, lies in the plane of the page.

A uniform magnetic field is directed vertically upwards (towards the top of the page), as shown in the diagram.

A steady current I enters the frame at corner P and leaves the frame at the diagonally opposite corner R .

Which statement correctly describes the initial motion of the frame immediately after the current is applied?

(Assume the frame is in free space and unaffected by gravity or external supporting forces.)



- A** The frame moves perpendicularly out of the plane of the page without rotating.
 - B** The frame moves perpendicularly into the plane of the page without rotating.
 - C** The frame moves towards the right of the page.
 - D** The frame moves towards the top of the page.
 - E** The frame rotates about a horizontal axis in the plane of the page.
 - F** The frame rotates about a vertical axis in the plane of the page.
 - G** The frame rotates about an axis perpendicular to the plane of the page in a clockwise direction.
 - H** The frame remains completely stationary.
- 13** A heater supplies power at a constant rate of 50 W to a 2.0 kg block of material. The specific heat capacity of the material is $500 \text{ J kg}^{-1} \text{ K}^{-1}$.
- Over a period of 200 s, the temperature of the block increases by 8.0 K.
- What is the average rate of energy loss from the block to the surroundings during this period?
- A** 10 W
 - B** 30 W
 - C** 40 W
 - D** 50 W
 - E** 90 W

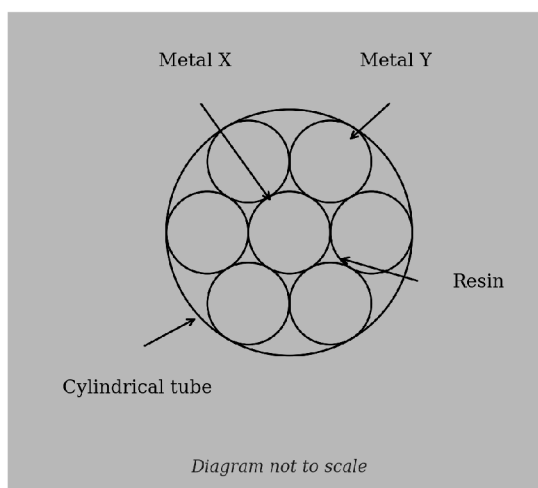
- 14** A progressive transverse wave travels along a horizontal rope with a speed of 12 m s^{-1} and a wavelength of 0.80 m . The amplitude of the wave is 2.5 cm .

What is the total distance travelled by a single point on the rope during the time it takes the wave to travel a distance of 24 m ?

- A** 0.75 m
B 1.5 m
C 3.0 m
D 6.0 m
E 24 m
- 15** A bundle of seven identical solid cylindrical rods, each of radius 1.0 cm and length 10.0 cm , is packed symmetrically inside a hollow cylindrical tube of internal radius 3.0 cm and length 10.0 cm , as shown in the cross-section below.

The central rod is made of metal X with density 8.0 g cm^{-3} , and the six outer rods are made of metal Y with density 3.0 g cm^{-3} . The remaining space inside the tube is completely filled with a resin of density 1.5 g cm^{-3} .

What is the total mass of the contents inside the tube?



- A** $240\pi \text{ g}$
B $260\pi \text{ g}$
C $290\pi \text{ g}$
D $305\pi \text{ g}$
E $340\pi \text{ g}$
F $540\pi \text{ g}$

- 16** A stationary radioactive nucleus decays, emitting an alpha particle of mass m_α and leaving a daughter nucleus of mass m_d . The kinetic energy of the emitted alpha particle is K .

What is the kinetic energy of the recoiling daughter nucleus?

(Assume non-relativistic speeds.)

- A** 0
- B** K
- C** $K \frac{m_d}{m_\alpha}$
- D** $K \frac{m_\alpha}{m_d}$
- E** $K \frac{m_\alpha}{m_d + m_\alpha}$
- F** $K \frac{m_d}{m_d + m_\alpha}$

- 17** A laser beam reflects from a plane mirror M_1 onto a second plane mirror M_2 , and then emerges in a final direction. The path of the beam lies within a 2D plane. Initially, both mirrors are stationary. M_1 is then rotated by an angle θ in a clockwise direction. The incident beam direction remains fixed.

To keep the final direction of the beam emerging from M_2 unchanged, what rotation must be applied to M_2 ?

- A** θ clockwise
- B** θ anticlockwise
- C** 2θ clockwise
- D** 2θ anticlockwise

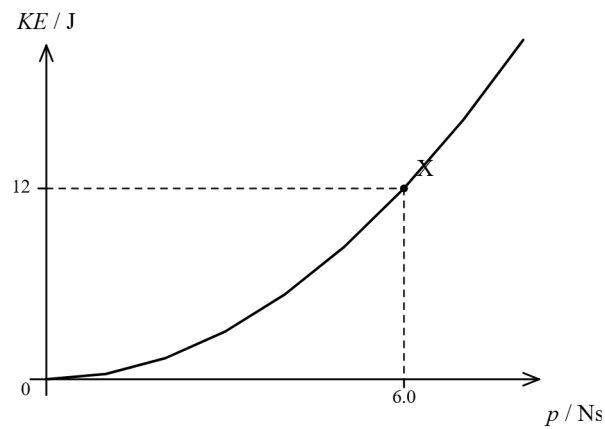
- 18** A block with a total volume of 1.0 m^3 is constructed from two materials, A and B. The density of material A is 200 kg m^{-3} and the density of material B is 1200 kg m^{-3} .

The block floats in water with exactly $\frac{4}{5}$ of its volume submerged. The density of water is 1000 kg m^{-3} .

What is the mass of material B in the block?

- A** 80 kg
- B** 600 kg
- C** 720 kg
- D** 800 kg
- E** 960 kg

- 19 The graph shows the kinetic energy, KE, of an object plotted against the magnitude of its momentum, p . A point X on the graph corresponds to a momentum of 6.0 Ns and a kinetic energy of 12 J.



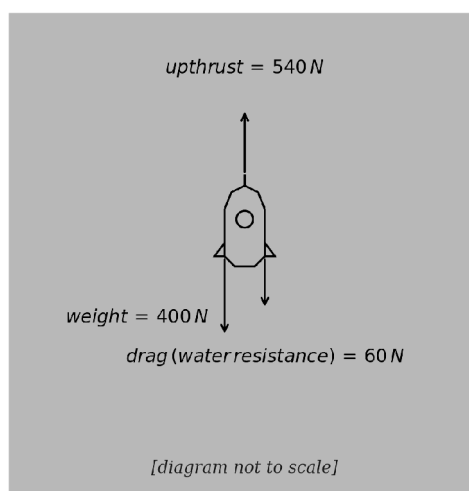
What is the speed of the object at this point?

- A 0.50 m s^{-1}
 - B 1.5 m s^{-1}
 - C 2.0 m s^{-1}
 - D 4.0 m s^{-1}
 - E 6.0 m s^{-1}
 - F 9.0 m s^{-1}
- 20 A uniform rectangular block of width 1.0 m and height 2.0 m rests on a rough horizontal surface. The coefficient of static friction between the block and the surface is μ . A horizontal force is applied to the top edge of the block. As the force is gradually increased, the block begins to tip over without sliding. Which inequality defines the possible values of μ ?
- A $\mu < 0.25$
 - B $\mu > 0.25$
 - C $\mu > 0.50$
 - D $\mu > 1.00$
 - E $\mu > 2.00$

- 21** An autonomous underwater probe of mass 40 kg is ascending vertically through water. At a particular instant, the vertical forces acting on the probe are shown in the diagram.

Which row in the table gives the magnitude and direction of the probe's acceleration at this instant?

	magnitude of acceleration / m s^{-2}	direction of acceleration
A	2.0	upwards
B	2.0	downwards
C	3.5	upwards
D	3.5	downwards
E	5.0	upwards
F	12.0	upwards



- 22** A uniform ladder of mass M and length L rests against a smooth vertical wall and a rough horizontal floor with coefficient of static friction μ . A person of mass M climbs the ladder.

Which of the following inequalities defines the range of angles θ (between the ladder and the floor) for which the ladder will remain stable for all positions of the person on the ladder?

- A** $\tan \theta \geq \frac{1}{4\mu}$
- B** $\tan \theta \geq \frac{1}{2\mu}$
- C** $\tan \theta \geq \frac{3}{4\mu}$
- D** $\tan \theta \geq \frac{1}{\mu}$

- 23** A detector is used to monitor the radiation emitted by a sample of a single radioactive isotope. The detector records the total count rate, which includes a constant background count rate.

The recorded count rate at three different times is:

- at $t = 0$ minutes, count rate = 420 counts per minute
- at $t = 40$ minutes, count rate = 120 counts per minute
- at $t = 80$ minutes, count rate = 45 counts per minute

Which row in the table gives the half-life of the isotope and the background count rate?

	Half-life / min	Background count rate / counts per min
A	20	15
B	20	20
C	20	25
D	40	15
E	40	20
F	40	25
G	80	20
H	80	45

- 24** A bat flies directly towards a wall at a constant speed of 5.0 m s^{-1} . When it is a distance d from the wall, it emits a short ultrasound pulse. The echo is received 0.120 s later. The speed of sound is 340 m s^{-1} .

What was the distance d when the pulse was emitted?

- A** 20.1 m
- B** 20.4 m
- C** 20.7 m
- D** 21.0 m
- E** 40.8 m
- F** 41.4 m

- 25** A singly positively charged ion P^+ of a particular element contains a total of 23 particles (protons, neutrons, and electrons).

The table shows information about the total number of particles and the relative charges of four atoms or ions W, X, Y, and Z.

atom or ion	total number of particles	relative charge
W	24	0
X	25	0
Y	26	-1
Z	22	+1

Which of these atoms or ions could be of a different isotope to P^+ but of the same element as P^+ ?

- A** X only
 - B** Y only
 - C** W and Z only
 - D** X and Y only
 - E** X and Z only
 - F** W, X and Y only
 - G** W, Y and Z only
 - H** X, Y and Z only
- 26** A wave travels 8.4 m in 0.70 s. The source produces 2.0 complete oscillations each second.
- What is the wavelength of the wave?
- A** 1.4 m
 - B** 2.4 m
 - C** 4.2 m
 - D** 6.0 m
 - E** 8.4 m
 - F** 12.0 m

- 27** A continuous progressive wave travels from Medium 1 into Medium 2. The speed of the wave decreases by 25% when it enters Medium 2. If the wavelength of the wave in Medium 1 is λ , what is the wavelength of the wave in Medium 2?

- A** $\frac{\lambda}{2}$
- B** $\frac{2\lambda}{3}$
- C** $\frac{3\lambda}{4}$
- D** λ
- E** $\frac{4\lambda}{3}$

Chemistry

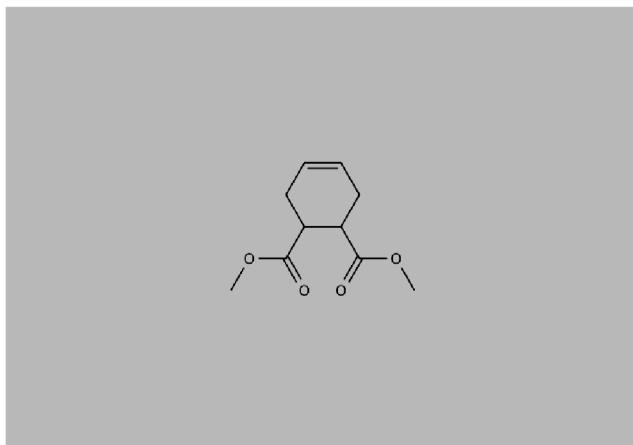
- 1 A sample of element ${}^{\text{X}}$ consists of only three isotopes: ${}^{68}\text{X}$, ${}^{70}\text{X}$, and ${}^{72}\text{X}$. The relative atomic mass (A_r) of ${}^{\text{X}}$ in this sample is 70.2.

isotope	percentage abundance / %
${}^{68}\text{X}$	x
${}^{70}\text{X}$	30
${}^{72}\text{X}$	y

What is the value of y ?

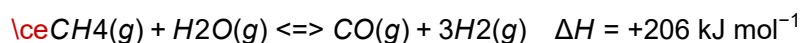
- A 20
B 30
C 35
D 40
E 45
- 2 Magnesium is an element in Group 2 of the Periodic Table.
Which of the following statements is/are correct about magnesium?
- 1 Magnesium is a stronger reducing agent than calcium.
 - 2 Magnesium has a smaller atomic radius than calcium.
 - 3 Magnesium reacts with nitrogen gas to form a compound containing 72% magnesium by mass.
- (A_r values: ${}^{\text{N}} = 14$; ${}^{\text{Mg}} = 24$; ${}^{\text{Ca}} = 40$)
- A none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

- 3 The skeletal formula of an organic compound is shown below.
What is the empirical formula of this compound?



- A $\text{C}_5\text{H}_7\text{O}_2$
 B $\text{C}_{10}\text{H}_{14}\text{O}_4$
 C $\text{C}_5\text{H}_6\text{O}_2$
 D $\text{C}_5\text{H}_8\text{O}_2$
 E $\text{C}_4\text{H}_7\text{O}_2$
 F $\text{C}_2\text{H}_3\text{O}$

- 4 The industrial production of synthesis gas from methane is represented by the following endothermic, reversible reaction:



A mixture of $\text{CH}_4(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ is placed in a closed vessel and allowed to reach dynamic equilibrium.

Which of the following statements is/are correct?

- 1 At dynamic equilibrium, the activation energy of the forward reaction is equal to the activation energy of the backward reaction.
- 2 Increasing the temperature at constant pressure increases the rate of both the forward and backward reactions, and increases the equilibrium yield of $\text{H}_2(\text{g})$.
- 3 Adding a catalyst decreases the time taken to reach equilibrium, but does not change the ratio $\frac{k_{\text{forward}}}{k_{\text{backward}}}$, where k_{forward} and k_{backward} are the rate constants for the forward and backward reactions.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 5 Two experiments were carried out using separate samples of an aqueous solution of an unknown base B of concentration $0.150 \text{ mol dm}^{-3}$ (11.1 g dm^{-3}), and the following observations were made:

* 20.0 cm^3 of the base solution was completely neutralised by 30.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ hydrochloric acid, HCl(aq) .

* A pale blue precipitate formed when a few drops of aqueous copper(II) sulfate, $\text{CuSO}_4(\text{aq})$, were added to a portion of the base solution.

Which of the following statements can be deduced from this information?

1. One mole of the base reacts with two moles of H^+ ions in complete neutralisation.
2. The base is fully dissociated in aqueous solution.
3. The relative formula mass (M_r) of the base is 74.

(A_r values are not required)

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3
- 6 What are the oxidation states of sulfur in SO_2 and SO_4^{2-} , respectively?
- A +2 and +4
- B +4 and +6
- C +4 and +4
- D +6 and +4
- E -4 and -6
- F 0 and +6

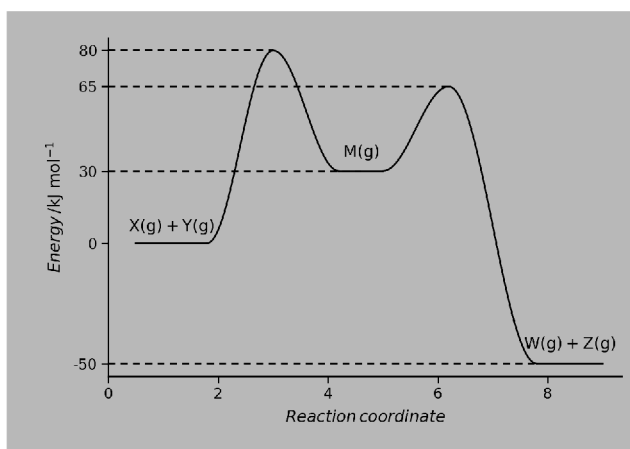
- 7 The diagram shows the reaction energy profile for the two-step gas-phase reaction:



where M is a reaction intermediate.

Which of the following statements is/are correct?

- 1 The rate-determining step for the forward reaction is the conversion of intermediate M into the products $W + Z$.
- 2 The activation energy for the overall reverse reaction is $+130 \text{ kJ mol}^{-1}$.
- 3 The conversion of intermediate M back into reactants has a higher activation energy than its conversion into products.

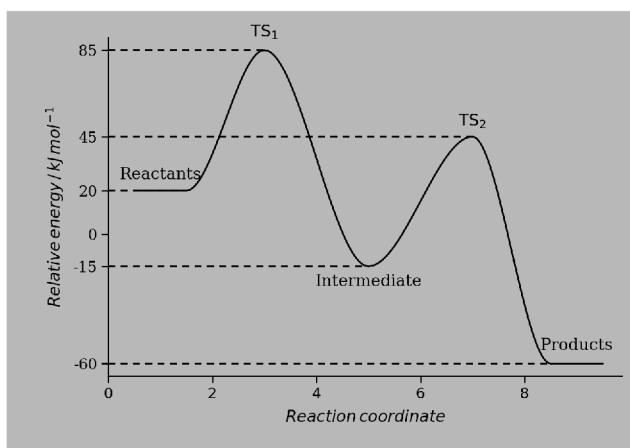


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 8 The energy profile for a two-step reaction, $\text{Reactants} \rightarrow \text{Intermediate} \rightarrow \text{Products}$, is shown in the diagram.

Which of the following statements is/are correct?

- 1 The overall reaction is exothermic with $\Delta H = -80 \text{ kJ mol}^{-1}$.
- 2 The first step ($\text{Reactants} \rightarrow \text{Intermediate}$) is the rate-determining step for the forward reaction.
- 3 The activation energy for the overall reverse reaction ($\text{Products} \rightarrow \text{Reactants}$) is 105 kJ mol^{-1} .



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 A diol reacts with an excess of propanoic acid in the presence of an acid catalyst to form a diester.

The diester has a relative molecular mass (M_r) of 174.

What is the relative molecular mass (M_r) of the diol?

(A_r values: $^1\text{H} = 1.0$; $^{12}\text{C} = 12.0$; $^{16}\text{O} = 16.0$)

- A 26
B 44
C 62
D 76
E 90
F 100
G 118
H 136
- 10 Researchers intend to isolate mitochondria from a liver cell homogenate using differential centrifugation. The table below displays the minimum gravitational force (g) required to pellet specific cellular components.

Component	Minimum Force to Pellet (g)
Nuclei	1,000
Mitochondria	15,000
Ribosomes	100,000

The homogenate is first centrifuged at 1,000 g for 10 minutes, producing a pellet and a supernatant.

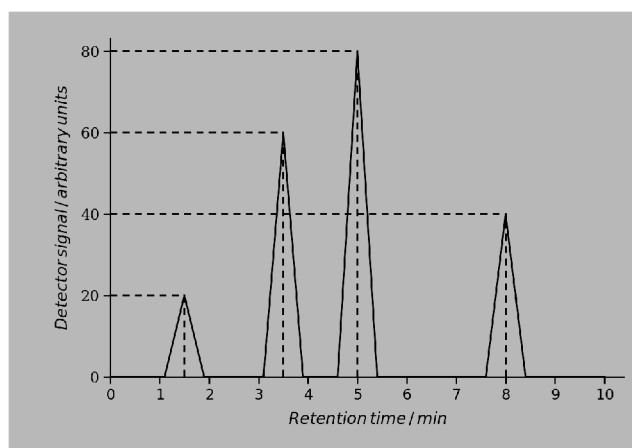
Which of the following subsequent steps would yield the purest sample of mitochondria?

- A Resuspend the pellet from the first step and centrifuge at 15 000 g.
B Centrifuge the supernatant from the first step at 15 000 g and collect the resulting pellet.
C Centrifuge the supernatant from the first step at 15 000 g and collect the resulting supernatant.
D Centrifuge the supernatant from the first step at 100 000 g and collect the resulting pellet.
E Centrifuge the supernatant from the first step at 100 000 g and collect the resulting supernatant.

- 11 A mixture containing four alkanes—two isomers with molecular formula C_5H_{12} (pentane and 2-methylbutane) and two alkanes with molecular formula C_6H_{14} (hexane and 2,2-dimethylbutane)—was analysed using gas-liquid chromatography with a non-polar stationary phase and helium carrier gas.

The resulting chromatogram is shown below. All four peaks have the same base width, and the detector response factor is identical for all four compounds.

What is the percentage by mass in the mixture of the C_5H_{12} isomer that has the strongest attraction to the stationary phase relative to the mobile phase?

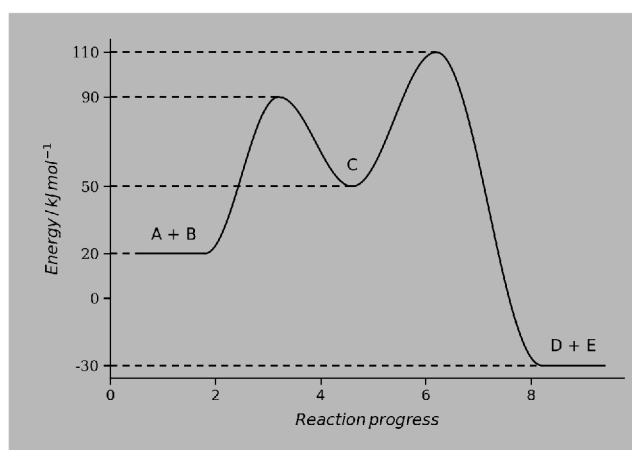


- A 10%
- B 20%
- C 30%
- D 40%
- E 60%
- F 75%

- 12 The reaction energy profile for the two-step reversible reaction $A + B \rightleftharpoons C \rightleftharpoons D + E$ is shown in the diagram.

Which row in the table gives the activation energy of the rate-determining step for the reverse reaction ($D + E \rightarrow A + B$) and the overall enthalpy change (ΔH) of the forward reaction?

	Activation energy of rate-determining step for reverse reaction / kJ mol^{-1}	Overall enthalpy change (ΔH) of forward reaction / kJ mol^{-1}
A	+40	-50
B	+60	+50
C	+90	-50
D	+140	-50
E	+140	+50
F	+160	-50

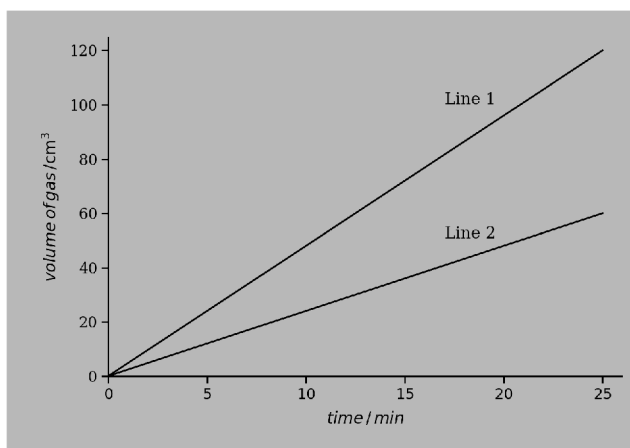


- 13** An aqueous solution of dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, is electrolysed at room temperature and pressure using inert platinum electrodes and a constant electric current.

The graph shows the volume of gas collected at each of the two electrodes as a function of time.

Which of the following statements is/are correct?

1. Line 1 represents the gas collected at the cathode.
2. In the external circuit, electrons flow from the electrode producing the gas of Line 1 towards the electrode producing the gas of Line 2.
3. Doubling the electric current would double the gradient of both Line 1 and Line 2.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

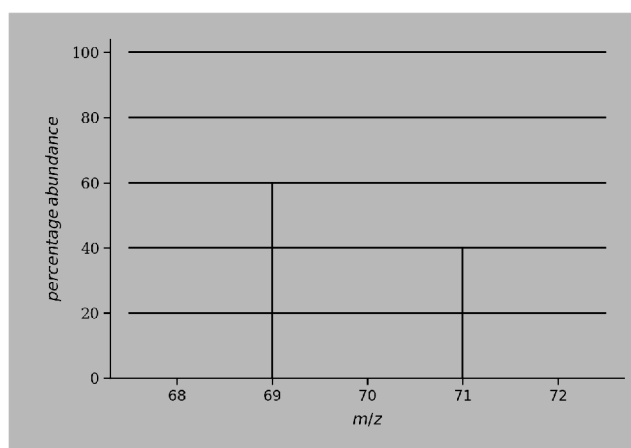
14 The mass spectrum of a sample of element M is shown in the diagram.

A 0.100 mol sample of M reacts completely with 2.40 g of oxygen gas (O_2) to form a single oxide.

Which row in the table gives the relative atomic mass of this sample of M and the empirical formula of the oxide formed?

($A_r(O) = 16.0$)

	Relative atomic mass	Empirical formula of oxide
A	69.8	MO
B	69.8	M_2O_3
C	69.8	MO_2
D	70.0	M_2O_3
E	70.0	MO_2
F	70.2	MO
G	70.2	M_2O_3
H	70.2	MO_2



15 Which of the following statements is/are correct?

- Ethene decolourises bromine water.
- Addition polymerisation of ethene produces a small-molecule by-product.
- Ethanol reacts with sodium to release hydrogen gas.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

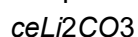
- 16** A solution contains both hydrochloric acid, HCl(aq) , and sulfuric acid, $\text{H}_2\text{SO}_4\text{(aq)}$.
- A 50.0 cm^3 sample of this solution is completely neutralized by 35.0 cm^3 of $0.200 \text{ mol dm}^{-3} \text{ NaOH(aq)}$.
 - A separate 50.0 cm^3 sample of the same solution is treated with excess aqueous barium chloride, $\text{BaCl}_2\text{(aq)}$, producing 0.466 g of dry $\text{BaSO}_4\text{(s)}$ precipitate.

What is the concentration of hydrochloric acid in the original solution?

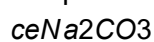
(A_r values: $\text{H} = 1.0$; $\text{O} = 16.0$; $\text{S} = 32.0$; $\text{Ba} = 137.0$)

- A** $0.020 \text{ mol dm}^{-3}$
 - B** $0.030 \text{ mol dm}^{-3}$
 - C** $0.040 \text{ mol dm}^{-3}$
 - D** $0.060 \text{ mol dm}^{-3}$
 - E** $0.080 \text{ mol dm}^{-3}$
 - F** $0.100 \text{ mol dm}^{-3}$
- 17** Three separate experiments are carried out. In each experiment, an excess of dilute hydrochloric acid is reacted with 0.050 mol of a solid Group 1 carbonate:

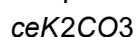
- Experiment 1:



- Experiment 2:



- Experiment 3:



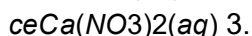
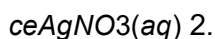
Let V_1 , V_2 , and V_3 be the final volumes of gas produced in Experiments 1, 2, and 3 respectively, measured at room temperature and pressure.

Which of the following correctly compares these volumes?

- A** $V_1 < V_2 < V_3$
 - B** $V_1 > V_2 > V_3$
 - C** $V_1 = V_2 = V_3$
 - D** $V_1 = V_2 < V_3$
 - E** $V_1 > V_2 = V_3$
- 18** The ions X^{2-} , Y^- and Z^{2+} are isoelectronic (they have the same electronic configuration).
- Which of the following statements about the **neutral atoms** X, Y and Z is correct?
- A** Element Z has the lowest atomic number.
 - B** Element Z is in the period immediately following elements X and Y.
 - C** Elements X, Y and Z are in the same period.
 - D** Element X is in Group 2 of the Periodic Table.
 - E** Element Y has a larger atomic radius than element Z.

- 19 Electrolysis is carried out on separate samples of the following four aqueous solutions using inert graphite electrodes:

1.



In which of these experiments is the metal deposited at the cathode?

- A 1 and 3 only
B 2 and 4 only
C 1, 2 and 3 only
D 3 and 4 only
E 1, 2, 3 and 4

- 20 The boiling points of the following three compounds are -100°C , -24°C , and 13°C (not necessarily in that order): 1. Boron trifluoride (BF_3) 2. Chloromethane (CH_3Cl) 3. Boron trichloride (BCl_3)

Which option lists the compounds in order of **increasing** boiling point? (Ar values: H = 1; B = 11; C = 12; F = 19; Cl = 35.5)

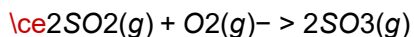
- A $\text{BF}_3 < \text{CH}_3\text{Cl} < \text{BCl}_3$
B $\text{BF}_3 < \text{BCl}_3 < \text{CH}_3\text{Cl}$
C $\text{CH}_3\text{Cl} < \text{BF}_3 < \text{BCl}_3$
D $\text{BCl}_3 < \text{CH}_3\text{Cl} < \text{BF}_3$

- 21 An element, M, is in Period 3 of the Periodic Table. The first four successive ionisation energies of M are 578 kJ mol^{-1} , 1817 kJ mol^{-1} , 2745 kJ mol^{-1} , and $11578 \text{ kJ mol}^{-1}$.

Which row in the table correctly identifies the formula of the oxide of M, the acid-base character of this oxide, and whether this oxide reacts with excess aqueous sodium hydroxide?

	formula of oxide	acid-base character of oxide	reaction with excess $\text{ceNaOH}(\text{aq})$
A	ceM_2O_3	amphoteric	reacts
B	ceM_2O_3	basic	does not react
C	ceM_2O_3	basic	reacts
D	ceM_2O_3	acidic	reacts
E	ceMO	basic	does not react
F	ceMO	amphoteric	reacts
G	ceMO_2	acidic	reacts
H	ceMO_2	amphoteric	reacts

- 22 The reaction energy profile for the oxidation of sulfur dioxide is shown below:

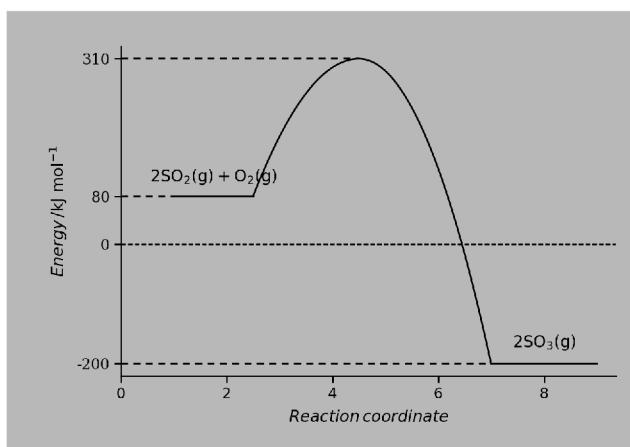


In an experiment, the thermal energy released by the complete oxidation of a sample of $\text{SO}_2(\text{g})$ is transferred to 600 g of water. The heat transfer process is 75% efficient.

The temperature of the water increases by 25°C .

What mass of SO_2 was reacted?

(A_r values: S = 32.0, O = 16.0. Specific heat capacity of water = $4.2 \text{ J g}^{-1}\text{C}^{-1}$)



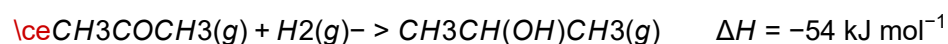
- A 10.8 g
B 19.2 g
C 21.6 g
D 38.4 g
E 48.0 g
F 53.8 g
- 23 The single-step gas-phase reaction $\text{ceX}(\text{g}) + 3\text{Y}(\text{g}) \rightleftharpoons 2\text{Z}(\text{g})$ is at equilibrium in a sealed container. The volume of the container is suddenly halved while the temperature is maintained at a constant value. What is the value of the ratio below, immediately after the change?

$$\frac{\text{rate of forward reaction}}{\text{rate of reverse reaction}}$$

- A $\frac{1}{4}$
B $\frac{1}{2}$
C 1
D 2
E 4
F 16

- 24 The structure of propan-2-ol is shown below:

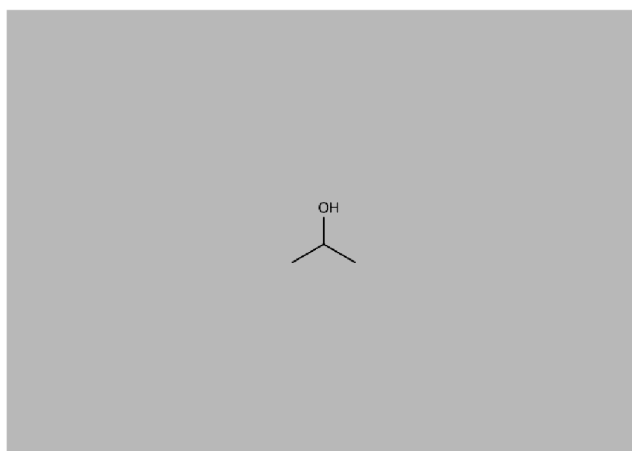
Propanone reacts with hydrogen gas to form propan-2-ol according to the following equation:



Using the mean bond enthalpies provided below, what is the mean bond enthalpy of the $\text{C}=\text{O}$ bond in propanone?

Mean bond enthalpies:

- $\text{C}-\text{C}$: 348 kJ mol^{-1}
- $\text{C}-\text{H}$: 413 kJ mol^{-1}
- $\text{C}-\text{O}$: 358 kJ mol^{-1}
- $\text{H}-\text{H}$: 436 kJ mol^{-1}
- $\text{O}-\text{H}$: 464 kJ mol^{-1}



- A 332 kJ mol^{-1}
 - B 440 kJ mol^{-1}
 - C 745 kJ mol^{-1}
 - D 853 kJ mol^{-1}
 - E 1181 kJ mol^{-1}
- 25 A chemical equation that represents the reaction of copper(I) phosphide with dilute nitric acid is:



where w , x , y , and z are integers.

What is the value of the sum $w + x + y + z$?

- A 49
- B 53
- C 57
- D 59
- E 63

- 26** When white phosphorus, P_4 , reacts with warm aqueous sodium hydroxide, it undergoes a disproportionation reaction to produce phosphine gas, PH_3 , and hypophosphite ions, H_2PO_2^- .

The unbalanced ionic equation for this reaction is:



What is the simplest whole-number ratio of $e : d$ in the balanced equation?

- A** 1 : 1
 - B** 1 : 2
 - C** 2 : 1
 - D** 1 : 3
 - E** 3 : 1
 - F** 1 : 4
 - G** 4 : 1
 - H** 3 : 4
- 27** An alkene **X** with the molecular formula C_6H_{12} exists as a pair of *E/Z* (geometric) isomers.

Addition polymerisation of **X** forms a polymer in which each repeating unit has:

- exactly one alkyl group attached to each carbon atom of the two-carbon backbone;
- two alkyl groups that contain the same number of carbon atoms as each other.

What is the IUPAC name of alkene **X**?

- A** hex-1-ene
- B** hex-2-ene
- C** hex-3-ene
- D** 2-methylpent-2-ene
- E** 3-methylpent-2-ene
- F** 4-methylpent-2-ene
- G** 2,3-dimethylbut-2-ene

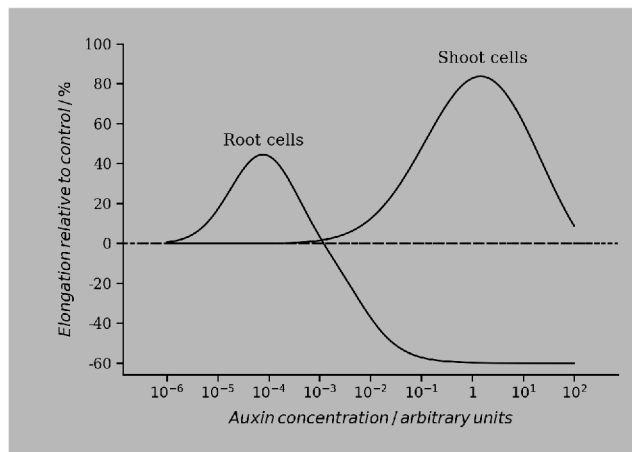
Biology

- 1 The graph shows the effect of auxin concentration on the elongation of root cells and shoot cells, expressed as the percentage change relative to an untreated control.

A germinating seedling is placed horizontally in the dark. Gravity causes auxin to accumulate on the lower side of both the root and the shoot, resulting in an auxin concentration of 10^{-1} arbitrary units on the lower side and 10^{-4} arbitrary units on the upper side in both organs.

Which of the following statements is/are correct?

- 1 The lower side of the shoot elongates faster than its upper side.
- 2 The lower side of the root elongates faster than its upper side.
- 3 The shoot will bend upwards and the root will bend downwards.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2 A bacterium and a typical animal cell are both undergoing asexual cell division. The bacterium contains a single circular DNA molecule, while the animal cell contains multiple linear chromosomes.

Which row in the table correctly describes the mechanism of genetic material separation in each cell?

	Bacterium	Animal Cell
A	Spindle fibres pull DNA to opposite poles	Spindle fibres pull DNA to opposite poles
B	Replicated DNA separates as the cell elongates	Spindle fibres pull sister chromatids apart
C	Replicated DNA separates as the cell elongates	Spindle fibres pull homologous chromosomes apart
D	Sister chromatids are pulled apart by spindle fibres	Spindle fibres pull homologous chromosomes apart
E	Homologous chromosomes are pulled apart by spindle fibres	Spindle fibres pull sister chromatids apart

- 3 During pregnancy in a mammal, exchange of substances occurs between maternal blood and fetal blood across the placenta.

Which row in the table correctly identifies the substances that are present at a higher concentration in the blood of the umbilical vein compared to the blood of the umbilical artery?

Key:

✓ = higher concentration in the umbilical vein

X = not a higher concentration in the umbilical vein

	glucose	carbon dioxide	urea
A	✓	✓	✓
B	✓	✓	X
C	✓	X	✓
D	✓	X	X
E	X	✓	✓
F	X	✓	X
G	X	X	✓
H	X	X	X

- 4 An investigation was carried out to study the rate of protein synthesis in two different strains of a yeast species, Strain X and Strain Y.

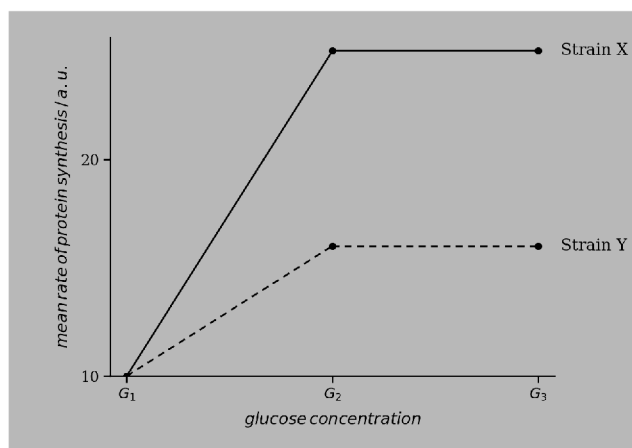
Within each strain, all yeast cells were genetically identical clones.

Samples containing equal numbers of cells from each strain were grown in nutrient solutions with different concentrations of glucose (G_1 , G_2 , and G_3 , where $G_1 < G_2 < G_3$). All other environmental conditions (temperature, pH, dissolved oxygen, and amino acid concentrations) were kept constant and identical for all cultures.

After a set period of time, the mean rate of protein synthesis per cell was measured for each culture. The results are shown in the graph.

Which of the following statements is/are correct?

- 1 The difference in the mean rate of protein synthesis in Strain X between G_1 and G_2 is due to environmental variation.
- 2 The difference in the mean rate of protein synthesis between Strain X and Strain Y at G_2 is due to genetic variation.
- 3 The failure of the mean rate of protein synthesis to increase in Strain X between G_2 and G_3 could be caused by an environmental factor other than glucose becoming limiting.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

5 Which of the statements about somatic gene therapy used to treat a human genetic disorder caused by a faulty recessive allele are correct?

- 1 A vector is used to introduce a functional copy of the allele into target body cells.
- 2 The introduced allele is incorporated into the genomes of both somatic cells and gametes.
- 3 The functional protein is synthesised using the ribosomes of the patient's own cells.
- 4 Successful treatment ensures that the patient cannot pass the faulty allele to any biological offspring.

- A** 1 and 2 only
B 1 and 3 only
C 2 and 4 only
D 3 and 4 only
E 1, 2 and 3 only
F 1, 3 and 4 only
G 2, 3 and 4 only
H 1, 2, 3 and 4

6 Three different mechanisms of transport across a cell membrane, X, Y and Z, are compared in the table.

Mechanism of membrane transport	Transports substances across a cell membrane	Requires transmembrane protein carriers or channels	Can result in net movement against a concentration gradient	Requires energy directly from ATP hydrolysis
X	yes	yes	no	no
Y	yes	no	no	no
Z	yes	yes	yes	yes

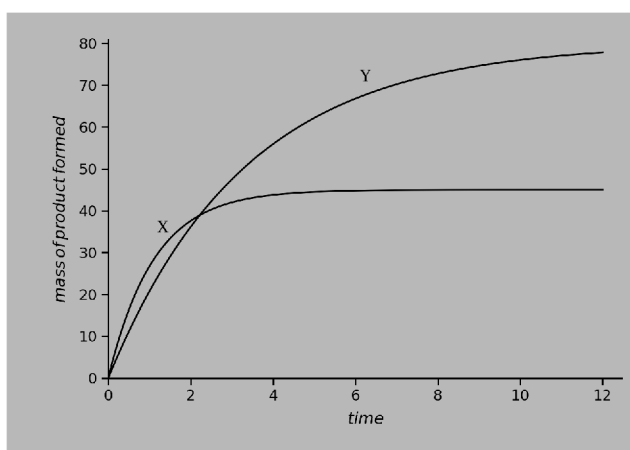
Which row in the table correctly identifies mechanisms X, Y and Z?

	X	Y	Z
A	facilitated diffusion	simple diffusion	active transport
B	facilitated diffusion	active transport	simple diffusion
C	simple diffusion	facilitated diffusion	active transport
D	simple diffusion	active transport	facilitated diffusion
E	active transport	facilitated diffusion	simple diffusion
F	active transport	simple diffusion	facilitated diffusion

- 7 The graph shows the total mass of product formed over time during an enzyme-catalysed reaction carried out under two different conditions, X and Y.

Which of the following statements could explain the difference between curve X and curve Y?

- 1 Condition X had a higher enzyme concentration and a lower initial mass of substrate than Condition Y.
- 2 Condition X was at a higher temperature that caused progressive thermal denaturation of the enzyme over time, whereas Condition Y was at a lower temperature at which the enzyme remained stable.
- 3 Condition X contained a competitive inhibitor, whereas Condition Y contained no inhibitor.
- 4 Condition X was at a pH further from the enzyme's optimum pH than Condition Y.



- A 1 only
- B 2 only
- C 1 and 2 only
- D 1 and 3 only
- E 2 and 4 only
- F 3 and 4 only
- G 1, 2 and 3 only
- H 1, 2, 3 and 4

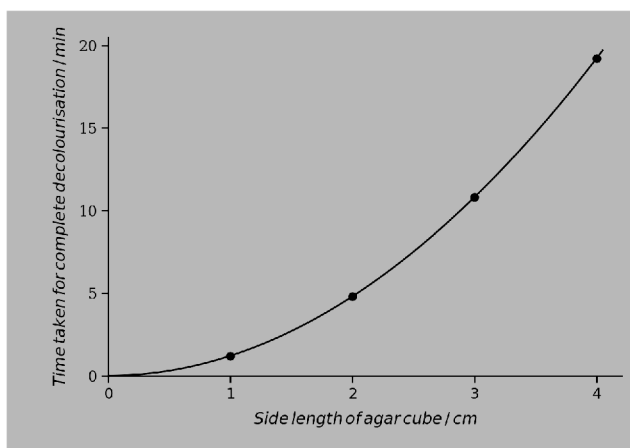
- 8 An investigation was carried out to model the effect of cell size on the rate of exchange of substances by diffusion.

Agar cubes of different side lengths were prepared containing dilute sodium hydroxide and the indicator phenolphthalein (which is pink in alkaline conditions). The cubes were placed into separate beakers containing equal volumes of dilute hydrochloric acid at the same concentration and temperature. The acid diffused into the cubes, neutralising the alkali and causing the cubes to decolourise.

The time taken for each cube to completely decolourise was recorded. The graph shows the results of this experiment.

Which of the following statements about the investigation is/are correct?

- 1 The dependent variable is plotted on the vertical (y) axis.
- 2 If the experiment were repeated at a higher temperature, the points plotted would lie below the original curve.
- 3 An agar block with dimensions $1\text{ cm} \times 2\text{ cm} \times 4\text{ cm}$ would take a longer time to completely decolourise than the $2\text{ cm} \times 2\text{ cm} \times 2\text{ cm}$ cube.



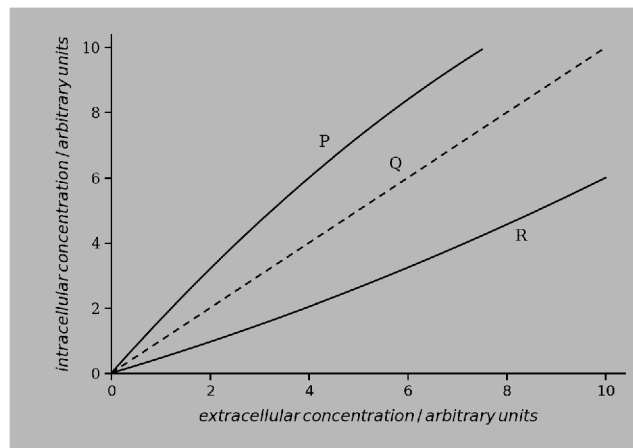
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 A cell is placed in solutions containing different concentrations of three uncharged solutes: P, Q, and R. The cell membrane is permeable to all three solutes.

The graph shows the intracellular concentration of each solute once a steady state is reached, as a function of the extracellular concentration. The dashed line represents where intracellular concentration equals extracellular concentration.

Which of the following statements is/are correct?

- 1 Metabolic energy (ATP) is required to maintain the steady-state concentration of solute P.
- 2 Metabolic energy (ATP) is required to maintain the steady-state concentration of solute R.
- 3 If ATP synthesis in the cell is inhibited, the steady-state intracellular concentration of solute R will decrease.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 10** Two mice, both heterozygous for two independently assorting autosomal genes ($AaBb$), are crossed. Under standard Mendelian inheritance with complete dominance, the expected phenotypic ratio among their offspring is 9 : 3 : 3 : 1.

However, across a very large number of offspring from these crosses, four distinct phenotypes are consistently observed in a ratio of 6 : 3 : 2 : 1.

Which of the following statements, if true, could explain this observed phenotypic ratio?

- 1 Homozygous dominant individuals for gene A (AA) die before birth, while Aa and aa individuals are viable.
- 2 Homozygous recessive individuals for gene A (aa) die before birth, while AA and Aa individuals are viable.
- 3 Homozygous dominant individuals for gene B (BB) die before birth, while Bb and bb individuals are viable.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 11** A 20-nucleotide synthetic mRNA molecule is translated in a cell-free translation system. Translation of this mRNA produces a hexapeptide (a peptide consisting of six amino acids) using one continuous reading frame from the 5' to 3' direction without any stop codons.

The sequence of the mRNA is shown below:

5'-AUUGGCAUUGACCGCAUUGC-3'

The table shows the number of each amino acid found in the resulting hexapeptide:

Amino acid	Number of residues
Leu	3
Pro	2
Met	1

What is the primary structure of the hexapeptide from the N-terminus to the C-terminus?

- A** Leu-Pro-Leu-Met-Pro-Leu
- B** Leu-Pro-Met-Leu-Pro-Leu
- C** Pro-Leu-Pro-Met-Leu-Leu
- D** Leu-Leu-Pro-Met-Pro-Leu
- E** Met-Leu-Pro-Leu-Pro-Leu
- F** Met-Pro-Leu-Pro-Leu-Leu
- G** Leu-Pro-Leu-Pro-Met-Leu
- H** Pro-Leu-Leu-Met-Leu-Pro

12 An experiment was conducted to investigate the gravitropic response of young primary roots.

Three identical batches of seedlings were placed with their primary roots oriented horizontally (0°) at time $t = 0$ h in humid, dark chambers under the following conditions:

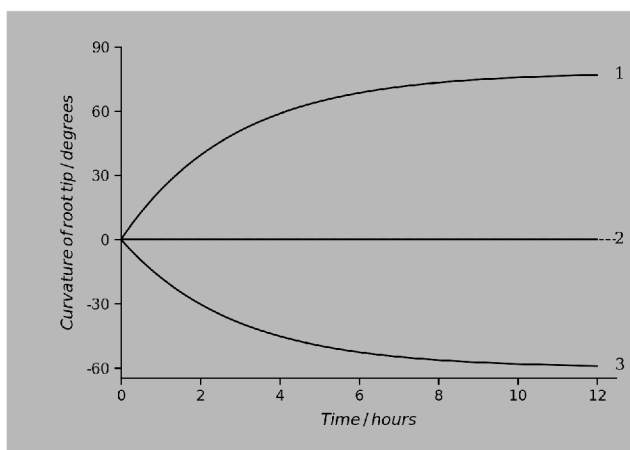
- **Treatment P:** Intact roots with no modifications (control).
- **Treatment Q:** Roots with the root cap surgically removed immediately before placement.
- **Treatment R:** Intact roots with an agar block containing a high concentration of auxin (IAA) placed only on the upper surface of the elongation zone.

The curvature of the root tips was recorded over a period of 12 hours. On the graph:

- a positive angle represents downward bending (towards gravity)
- a negative angle represents upward bending (away from gravity)
- an angle of 0° represents straight horizontal growth.

Which row of the table correctly matches each treatment to its corresponding curve on the graph?

	P	Q	R
A	1	2	3
B	1	3	2
C	2	1	3
D	2	3	1
E	3	1	2
F	3	2	1
G	1	2	1



- 13** An experiment was carried out in total darkness to investigate the gravitropic responses of young seedlings. Three identical, straight seedlings were placed horizontally at time $t = 0$ h under the following conditions:

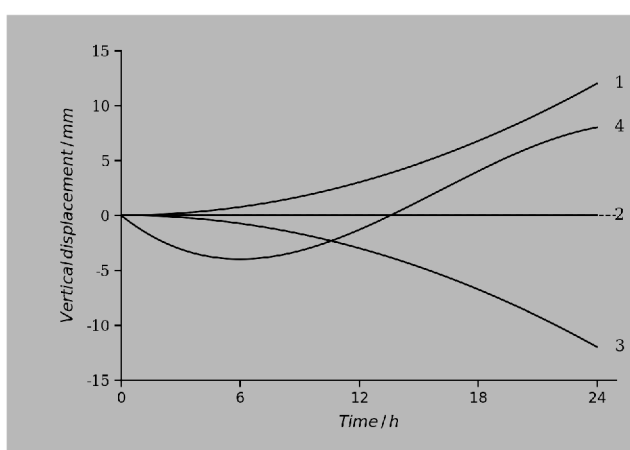
- **P**: An intact seedling kept stationary in a horizontal position; the vertical displacement of the shoot tip from the initial horizontal axis was measured over 24 hours.
- **Q**: An intact seedling kept stationary in a horizontal position; the vertical displacement of the root tip from the initial horizontal axis was measured over 24 hours.
- **R**: An intact seedling placed horizontally on a clinostat rotating slowly and continuously around its horizontal axis; the vertical displacement of the shoot tip from the initial horizontal axis was measured over 24 hours.

For all measurements, an upward displacement from the initial horizontal axis is defined as positive, and a downward displacement is defined as negative.

The graph shows four possible curves (1, 2, 3, and 4) representing vertical displacement over time.

Which row in the table correctly identifies the curves corresponding to seedlings P, Q, and R?

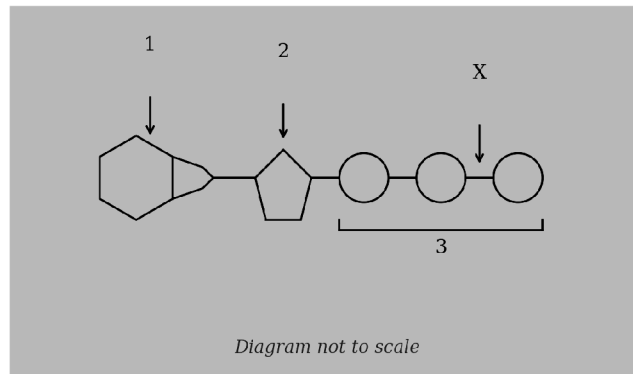
	P	Q	R
A	1	2	3
B	1	3	1
C	1	3	2
D	2	1	3
E	2	3	2
F	3	1	2
G	3	1	3
H	3	2	1



- 14** The diagram shows the structure of an adenosine triphosphate (ATP) molecule with components labelled 1, 2, and 3, and a specific bond labelled X.

Which of the following statements is/are correct?

- 1 Components 1 and 2, when bonded to a single phosphate group, form an adenine RNA nucleotide.
- 2 Hydrolysis of bond X produces adenosine diphosphate (ADP) and an inorganic phosphate group (P_i).
- 3 Component 2 is a pentose sugar containing one fewer oxygen atom than the pentose sugar found in DNA nucleotides.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 15** The table below shows the mean rates of water uptake by roots and water loss by transpiration for a leafy plant over consecutive 2-hour intervals between 06:00 and 18:00 on a warm day.

Time interval	Mean rate of water uptake by roots / cm h^{-3-1}	Mean rate of water loss by transpiration / cm h^{-3-1}
06:00 – 08:00	15	10
08:00 – 10:00	25	40
10:00 – 12:00	40	65
12:00 – 14:00	50	55
14:00 – 16:00	45	30
16:00 – 18:00	25	10

Which of the following statements is/are correct?

- 1 The plant experiences its greatest internal water deficit (highest xylem tension) at 14:00.
- 2 By 18:00, the plant has fully recovered the water deficit accumulated between 08:00 and 14:00.
- 3 The water potential in the leaf mesophyll cells is lower at 14:00 than at 08:00.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 16** The table shows characteristics of six plant species (A–F) inhabiting a fragmented landscape undergoing rapid climate change.

Which species is most at risk of extinction?

Species	Generation time	Dispersal ability	Genetic diversity	Ecological niche
A	short	high	high	generalist
B	short	low	high	specialist
C	long	low	low	specialist
D	long	high	low	generalist
E	long	low	high	generalist
F	short	low	low	specialist

- 17** A segment of double-stranded DNA is transcribed to produce an mRNA molecule containing 600 nucleotides.

The base composition of this mRNA molecule is:

- Adenine (A): 24%
- Uracil (U): 18%
- Cytosine (C): 32%
- Guanine (G): 26%

Which of the following statements about this 600-base-pair DNA segment is/are correct?

- 1 There are 192 cytosine bases in the template DNA strand.
- 2 Thymine accounts for 21% of the total nitrogenous bases in the double-stranded DNA segment.
- 3 There are 1548 hydrogen bonds between the complementary base pairs in this double-stranded DNA segment.

- A** none
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 18** In a species of moth, sex is determined by the ZW sex-chromosome system: females are heterogametic (ZW) and males are homogametic (ZZ).

A breeding experiment begins with a single mated pair of moths (Generation 0).

In each generation:

- each female mates with a male from the same generation and produces a single clutch of 10 offspring;
- the expected sex ratio of male to female offspring in each clutch is 1 : 1, and this ratio is observed in all clutches;
- all offspring survive to maturity and reproduce only within their generation.

What is the expected total number of Z chromosomes present among all individuals in the 4th generation of offspring (Generation 4)?

- A** 375
- B** 625
- C** 1250
- D** 1875
- E** 2500
- F** 3125
- G** 3750

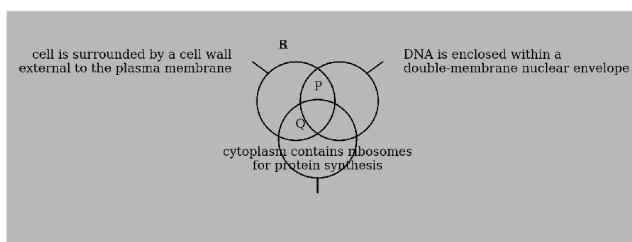
19 The Venn diagram shows three cellular features used to classify biological cells:

- Feature 1: Cell is surrounded by a cell wall external to the plasma membrane
- Feature 2: DNA is enclosed within a double-membrane nuclear envelope
- Feature 3: Cytoplasm contains ribosomes for protein synthesis

A bacterial cell, a yeast cell, and a human cheek epithelial cell can each be placed into one of the labelled regions P, Q, R, or S.

Which row in the table correctly identifies the region for each cell type?

	bacterial cell	yeast cell	human cheek epithelial cell
A	Q	S	R
B	Q	R	P
C	P	S	R
D	S	Q	P
E	S	S	R
F	R	S	Q
G	Q	P	R
H	P	Q	S



20 Root hair cells of a plant are placed in an aerated nutrient solution.

The concentrations of two dissolved ions, ion X and ion Y, and the water potential in the solution and in the root hair cell cytoplasm are:

- Ion X: concentration in solution = 0.1 mmol dm^{-3} ; concentration in cytoplasm = 4.5 mmol dm^{-3}
- Ion Y: concentration in solution = 8.0 mmol dm^{-3} ; concentration in cytoplasm = 1.2 mmol dm^{-3}
- Water potential: solution = -80 kPa ; cytoplasm = -450 kPa

Under normal aerated conditions, there is net movement of ion X, ion Y, and water into the root hair cells.

A chemical that specifically inhibits ATP synthesis is added to the nutrient solution.

Which row in the table shows the net movement of ion X, ion Y, and water across the root hair cell surface membrane immediately after the chemical is added?

	net movement of ion X	net movement of ion Y	net movement of water
A	moves into the cell	moves into the cell	moves into the cell
B	moves into the cell	moves into the cell	moves out of the cell
C	moves into the cell	does not move into the cell	moves into the cell
D	moves into the cell	does not move into the cell	moves out of the cell
E	does not move into the cell	moves into the cell	moves into the cell
F	does not move into the cell	moves into the cell	moves out of the cell
G	does not move into the cell	does not move into the cell	moves into the cell
H	does not move into the cell	does not move into the cell	moves out of the cell

- 21 The table shows several mRNA codons and the amino acids or translation functions they code for. All other codons not listed code for amino acids other than glutamic acid, aspartic acid, lysine, glutamine, or tryptophan, and none of the unlisted codons act as stop codons.

mRNA codon	Amino acid / function
GAA	glutamic acid
GAG	glutamic acid
GAC	aspartic acid
GAU	aspartic acid
AAG	lysine
CAG	glutamine
UAG	STOP
UGA	STOP
UAA	STOP
UGG	tryptophan

Consider the following six-base segment of an mRNA molecule:

5'-GAG UGG-3'

A single random base substitution mutation occurs within this six-base sequence. The mutation is equally likely to occur at any of the six nucleotide positions. At the mutated position, the original base is replaced by one of the other three RNA bases with equal probability ($\frac{1}{3}$ each).

Assuming no other mutations occur, what are the probabilities of the following outcomes?

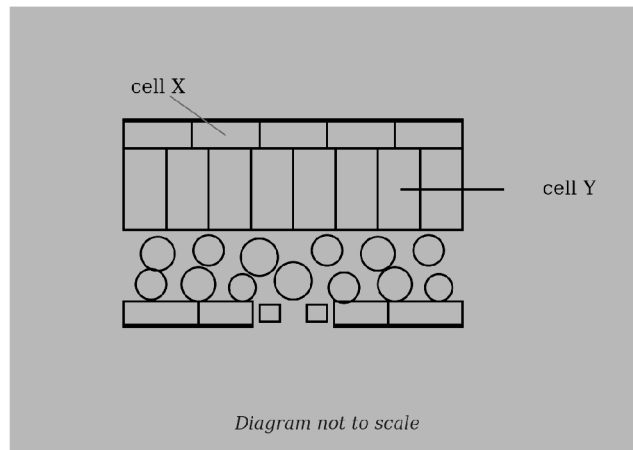
	Probability that the mutation is silent (both codons still code for glutamic acid followed by tryptophan)	Probability that the mutation produces a stop codon in this sequence
A	$\frac{1}{3}$	$\frac{1}{3}$
B	$\frac{1}{3}$	$\frac{2}{3}$
C	$\frac{1}{3}$	$\frac{2}{3}$
D	$\frac{1}{3}$	$\frac{2}{3}$
E	$\frac{2}{3}$	$\frac{1}{3}$
F	$\frac{2}{3}$	$\frac{2}{3}$
G	$\frac{2}{3}$	$\frac{2}{3}$
H	$\frac{2}{3}$	$\frac{2}{3}$

22 The diagram shows a simplified cross-section through part of a dicotyledonous leaf.

Consider the following statements about the two cells labelled X and Y:

- 1 Both cell X and cell Y belong to the same plant tissue.
- 2 Both cell X and cell Y contain the same nuclear genes.
- 3 Both cell X and cell Y contain chloroplasts.

Which of these statements is/are correct?



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 23** The table below displays annual carbon flux data for three different ecosystems, measured in arbitrary units of mass per area per year.

Ecosystem	Gross Primary Production (GPP)	Autotrophic Respiration (Ra)	Heterotrophic Respiration (Rh)
1	120	50	40
2	100	40	65
3	150	80	70

Which of the following conclusions regarding the carbon balance of these ecosystems is supported by the data?

- A** Ecosystem 3 acts as the largest net carbon sink because it has the highest Gross Primary Production.
- B** Ecosystem 2 acts as a net carbon source because its total respiration exceeds its Gross Primary Production.
- C** Ecosystems 1 and 3 contribute equally to the net removal of atmospheric carbon because their Net Primary Production is identical.
- D** Ecosystem 2 acts as a net carbon sink because the carbon fixed by photosynthesis exceeds the carbon released by producers.
- E** Ecosystem 1 is a net carbon source because the sum of autotrophic and heterotrophic respiration is highest relative to its GPP.

- 24** A marine biologist investigated the population of a burrowing bivalve species on a large intertidal mudflat.

A square core sampler with internal base dimensions of $0.20 \text{ m} \times 0.20 \text{ m}$ was pushed into the sediment to a uniform depth of 0.15 m . The sediment from the core was sieved to count the number of live bivalves present. This procedure was repeated 10 times.

Due to deep, soft mud further out on the mudflat, all 10 core samples were taken within 2 m of the high-tide line along the upper shore.

The number of bivalves recorded in each core is shown in the table:

Sample number	1	2	3	4	5	6	7	8	9	10
Number of bivalves	3	7	0	5	0	4	8	2	6	5

The biologist made the following statements:

- 1 Each core sampled a sediment volume of $6.0 \times 10^{-3} \text{ m}^3$.
- 2 The frequency of occurrence of the bivalves in the samples was 80%.
- 3 An accurate estimate of the total bivalve population on the entire mudflat could be calculated using these data if the total surface area of the mudflat was known.

Which of the statements about the investigation is/are correct?

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

25 In domestic cats, fur colour is controlled by an X-linked gene with two codominant alleles:

- * allele X^B produces black fur
- * allele X^O produces orange fur
- * heterozygous females ($X^B X^O$) have tortoiseshell fur

Two tortoiseshell female cats each mate with a different male cat:

- * Female 1 mates with a black male cat ($X^B Y$)
- * Female 2 mates with an orange male cat ($X^O Y$)

Each mating produces an equal number of male and female offspring, and both litters contain the same total number of kittens.

What is the expected proportion of all female kittens that are tortoiseshell, and what is the expected proportion of all kittens (males and females combined) across both litters that are orange?

	expected proportion of female kittens that are tortoiseshell	expected proportion of all kittens that are orange
A	0.25	0.250
B	0.25	0.375
C	0.25	0.500
D	0.50	0.250
E	0.50	0.375
F	0.50	0.500
G	0.75	0.375
H	0.75	0.500

26 A linear double-stranded DNA molecule contains 3000 base pairs and has a total of 7800 hydrogen bonds between its paired bases.

Which of the following statements is/are correct?

- 1 Guanine accounts for 60% of the bases in this DNA molecule.
- 2 There are 1200 adenine bases present in this DNA molecule.
- 3 A single strand of this DNA molecule contains 2999 phosphodiester bonds.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 27 The table shows the partial pressures of oxygen (p_{ceO_2}) and carbon dioxide (p_{ceCO_2}), and the concentration of urea, in blood entering and leaving three different organs (X, Y, and Z) of a healthy human at rest:

Organ	p_{ceO_2} entering / kPa	p_{ceO_2} leaving / kPa	p_{ceCO_2} entering / kPa	p_{ceCO_2} leaving / kPa	Urea entering / mmol dm ⁻³	Urea leaving / mmol dm ⁻³
X	13.0	5.5	5.2	6.0	5.0	2.5
Y	5.5	13.0	6.0	5.2	5.0	5.0
Z	13.0	5.5	5.2	6.0	5.0	7.5

Which row in the table below correctly identifies organs X, Y, and Z?

	Organ X	Organ Y	Organ Z
A	kidney	lungs	liver
B	kidney	liver	lungs
C	lungs	kidney	liver
D	lungs	liver	kidney
E	liver	kidney	lungs
F	liver	lungs	kidney